Lauren Kimpel

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Education

2024 - present PhD in Mathematics, **The University of Delaware**
2021 - 2024 M.S. in Applied & Computational Mathematics, **Johns Hopkins University**
2016 - 2020 B.S. in Mathematics & Computer Science, **Temple University**

Research

Banaian, Esther et al. (2024). "The e-positivity of the chromatic symmetric function for twinned paths and cycles". In: *Discrete Mathematics*. URL: https://api.semanticscholar.org/CorpusID:270067954.

Kimpel, Lauren (2024). "Some Fellow-Traveler Properties on Finite Graphs". MA thesis. Johns Hopkins University.

Cyber Applied Research Scientist Jan 2021 - Aug 2024
Lockheed Martin Advanced Concepts Laboratory Cherry Hill, NJ

- Played a key role in developing a novel network optimization and reasoning framework, resulting in an accepted research proposal to a U.S. DoD customer, and securing funding for a multi-year program
- Performed static and dynamic reverse engineering on x86/x86_64, SPARC, and PowerPC architectures. Mapped memory & control flow. Developed prototypes for cyber capabilities.
- Responsible for literature reviews and whitepaper writeups.
- Prototyped solutions related to integer programming, limited-information reasoning and symbolic execution. Wrote Python/C simulations.

Undergraduate Research Assistant May - Sep 2020
Computer & Information Sciences Department, Temple University Philadelphia, PA

- Analyzed the Pufferfish differential privacy framework for concrete tradeoffs between privacy levels and data freshness when sanitizing sensitive time series data
- Developed 2-state Markov chain-based mechanisms to probabilistically account for vulnerabilities related to data freshness metrics in ϵ-Pufferfish Private algorithms

Teaching

Graduate Teaching Assistant Aug 2024 - August 2025
Department of Mathematical Sciences, University of Delaware Newark, DE

- MATH 231 (Integrated Calculus 1A), Fall 2025
- MATH 115 (Precalculus), Fall 2025
- MATH 205 (Statistical Methods); Spring 2025
- MATH 241 (Analytic Geometry and Calculus A); Fall 2024

Undergraduate Teaching Assistant Aug - Dec 2020
Computer & Information Sciences Department, Temple University Philadelphia, PA

- CIS 3223 (Data Structures & Algorithms); Fall 2020

Conferences and Workshops

Graphical Models in Algebraic Combinatorics Summer School June - July 2025
Simons Laufer Mathematical Sciences Institute Berkeley, CA
Graduate Research Workshop in Combinatorics July 2023
The University of Wyoming Laramie, WY

Projects

Puffin: Simplicial Path Complexes for Obfuscation Detection 2023

- Transformed the control flow graphs (CFGs) of 12 binaries into 3-simplicial path complexes, using a filtration distance based on the combinatorial distance metric (maximum-length shortest path) between two vertices.
- Binaries were obfuscated by inserting extra bytes into the assembly code (`.s` files.) These are interpreted by the assembler to be extra `JMP` instructions, which causes the CFG to admit a more complex structure.
- Analyzed the homological differences between the unobfuscated and obfuscated binaries, and used the resulting data to train a Random-Forest Classifier to identify, given an arbitrary binary, whether it had been obfuscated using the `JMP`-insertion technique.

Mentorship

Graduate Mentor September-December 2025
University of Delaware Directed Reading Program Newark, DE

- Responsible for regularly giving suggested readings and research guidance to a second-year undergraduate student in the field of combinatorics, and, where applicable, coming up with interesting and challenging problems appropriate to the background(s) of both mentor and mentee, culminating in a research project and presentation.

Volunteering

Member, Association for Women in Mathematics (Student Chapter) 2025
Presenter, Young Women's Conference in Science, Engineering, & Mathematics 2023
Princeton University

Other Experience

Multidisciplined Linguist Intern May - Aug 2019
The United States Department of Defense Fort Meade, MD

- Obtained Top Secret security clearance and passed lifestyle polygraph
- Analyzed Russian morphologic, syntactic and phonetic structures in graphical and voice material to determine subtext, meaning and intention for high priority analysis and production mission critical to IC community customers, including senior U.S. decisionmakers

Skills

Programming Languages	Python, C, R, MATLAB, LaTeX, Java
Assembly	x_86/x_86-64, PowerPC
Reverse Engineering & Debugging Tools	Ghidra, WinDbg, gdb, Wireshark
Data Visualization & Analysis	networkX, matplotlib
Foreign Languages	English (native), Russian[1]
Other	Atlassian Tools, Git, Windows, Linux, Emacs, SageMath

Last updated: October 23, 2025

[1] can read on a basic level and write on a slightly more intermediate level; can speak but with nontrivial levels of impairment.